Do Now · retrieval

Lesson 153 · Integrated Web Forms: HTML, CSS, JavaScript and Server Checks

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which property writes safe plain text in DOM?

Q2

What is a database foreign key?

Q3

What is the role of a router?

Q4

What does a cache hit mean?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Capture — HTML labels and input types structure a form.

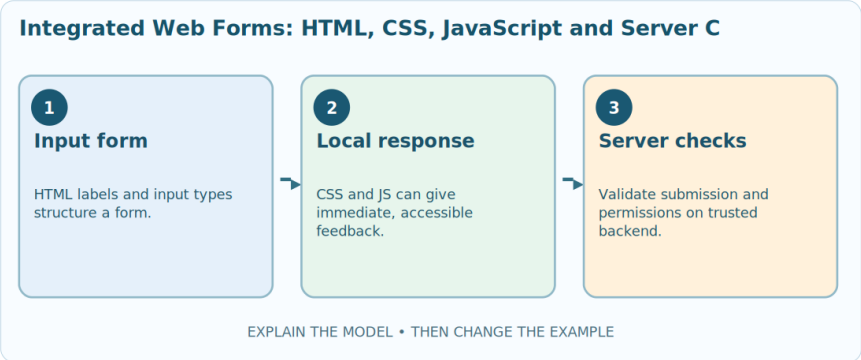
Core idea 2

Respond — CSS and JS can give immediate, accessible feedback.

Core idea 3

Trust the server — Validate submission and permissions on trusted backend.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

client-side — Execution on user device.

server-side — Execution on controlled backend.

validation — Checking input against rules.

trust boundary — Point where untrusted data enters a protected system.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

An online form permits Years 12 or 13. Where must that rule be checked?

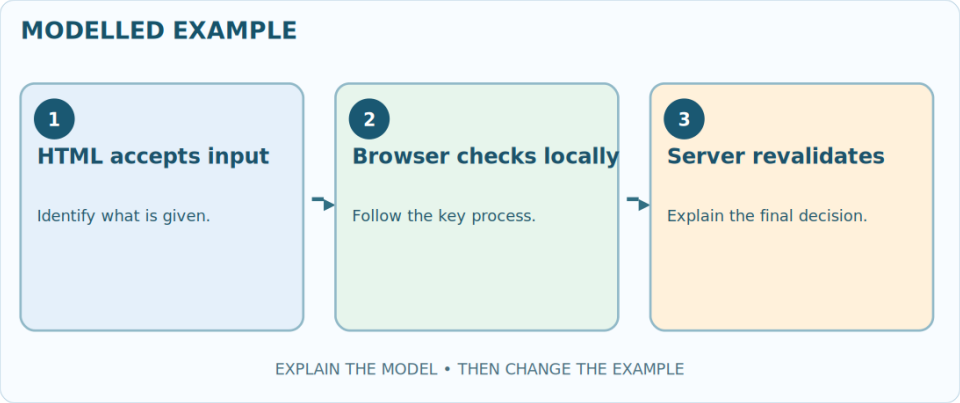
Follow the worked method

1. HTML accepts input — Identify the input or diagram and predict the first result.

2. Browser checks locally — Follow the second stage and explain its importance.

3. Server revalidates — HTML min/max and JavaScript improve normal experience; the server must check the supplied value independently. Client code can be changed or bypassed; no database update should trust it.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Client total altered
Identify the input or diagram and predict the first result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Distinguish roles of HTML, CSS and JavaScript in a form. [6]

Question 2

Explain why HTML required alone does not protect database data quality. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Design an accessible 16–19 age registration flow with client and server validation. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A number input with min/max means a malicious client cannot submit other values.

Use key vocabulary from memory

1. Explain client-side in your own words.

2. Explain server-side in your own words.

3. Explain validation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.